

On Parameter Loci of the Hénon Family (Joint with Yutaka Ishii)

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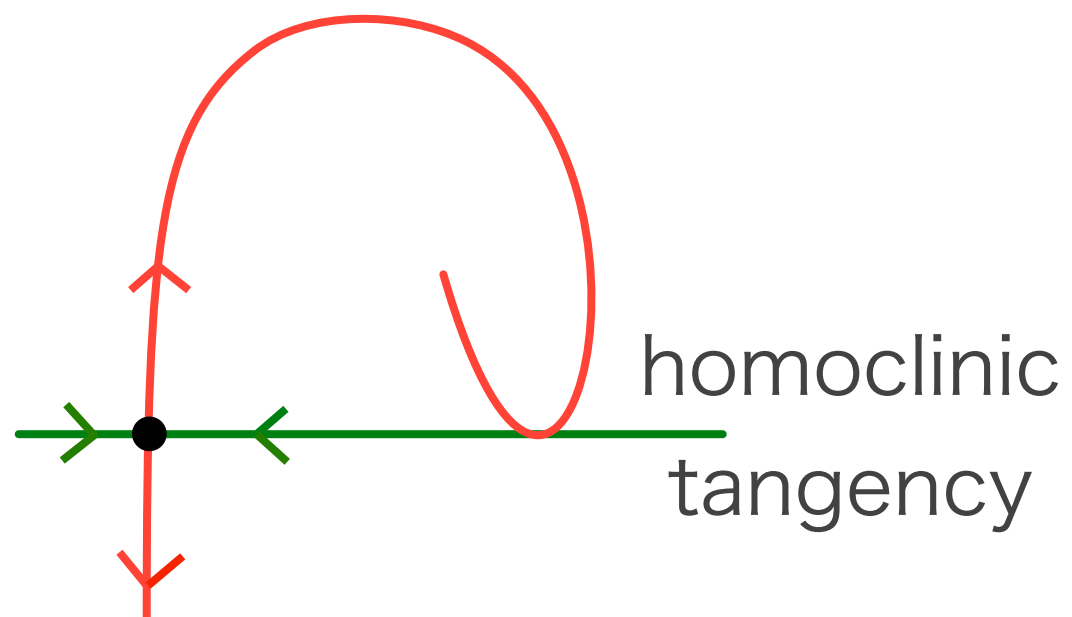
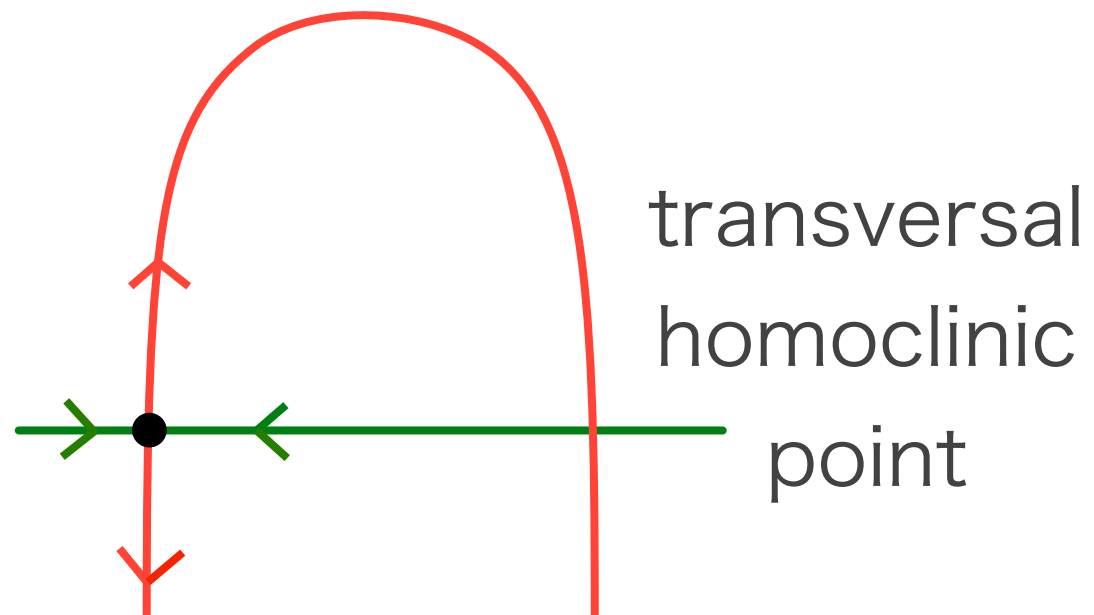
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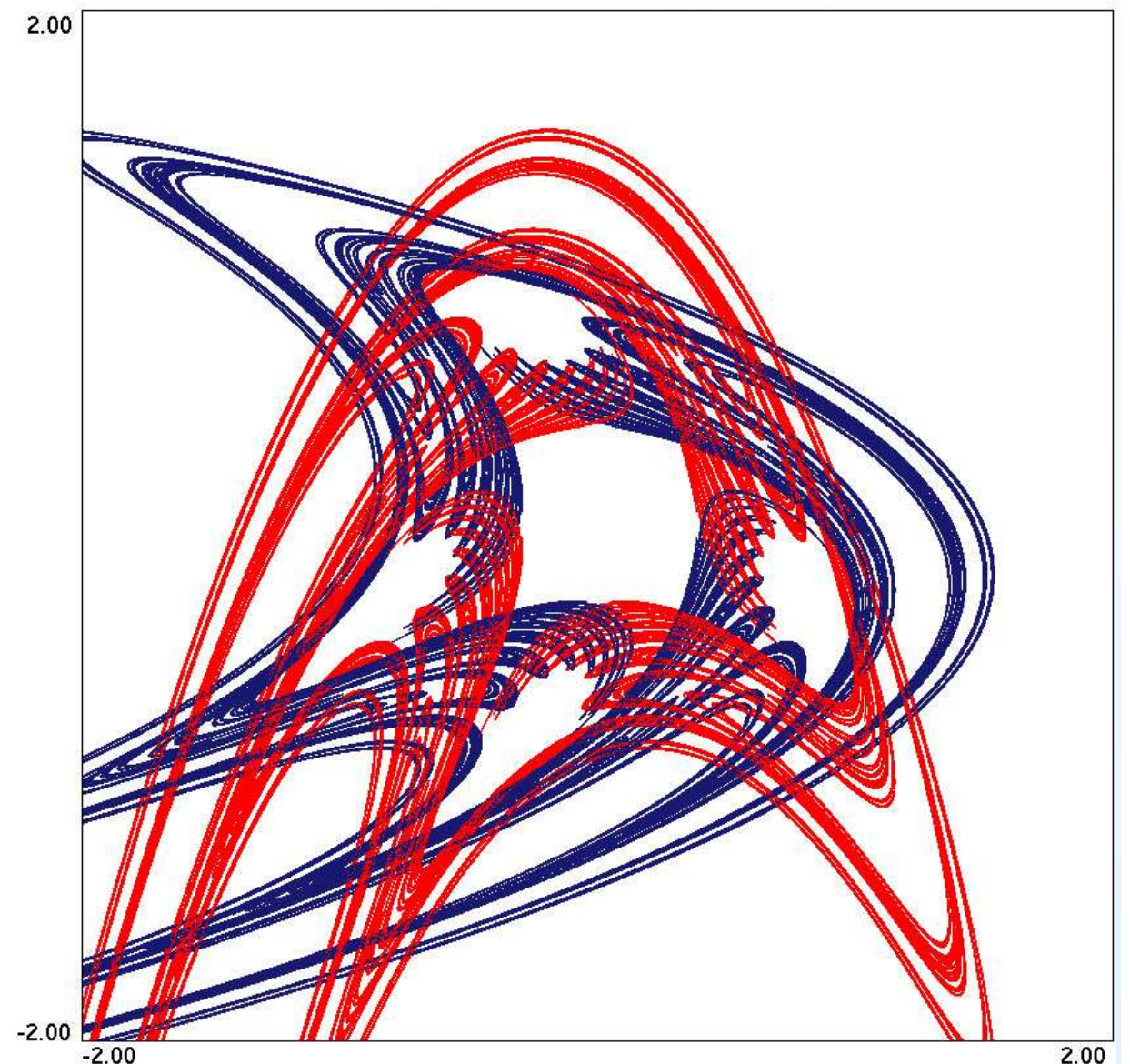
Background and Main Results

Homoclinic Points

Stable and unstable manifolds may have intersections.



$$F(x, y) = (0.1 - x^2 - y, x)$$



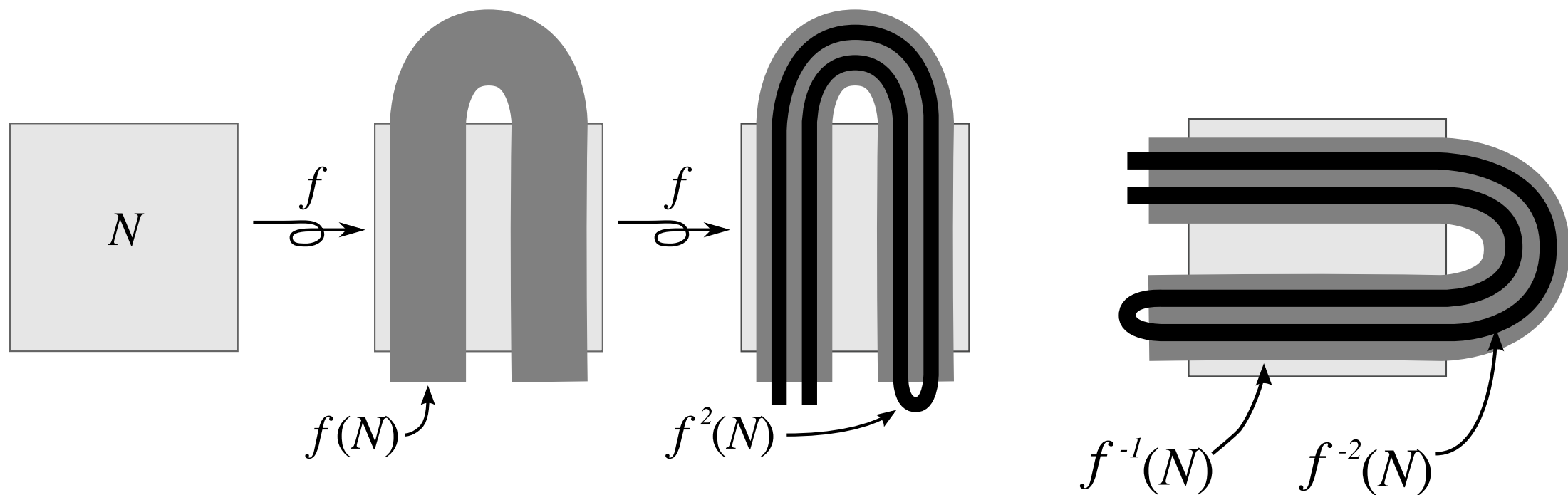
Smale Horseshoe



Stephan Smale

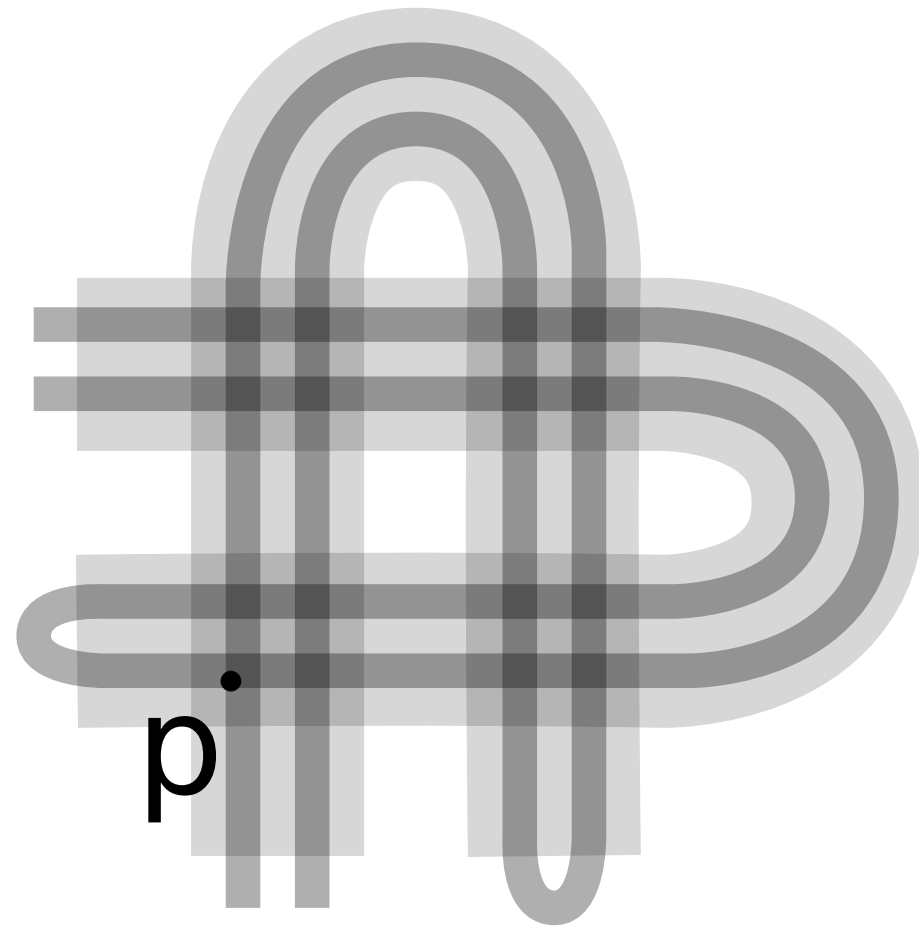


“Finding a horseshoe on the beaches of Rio”



Horseshoe and Homoclinic Points

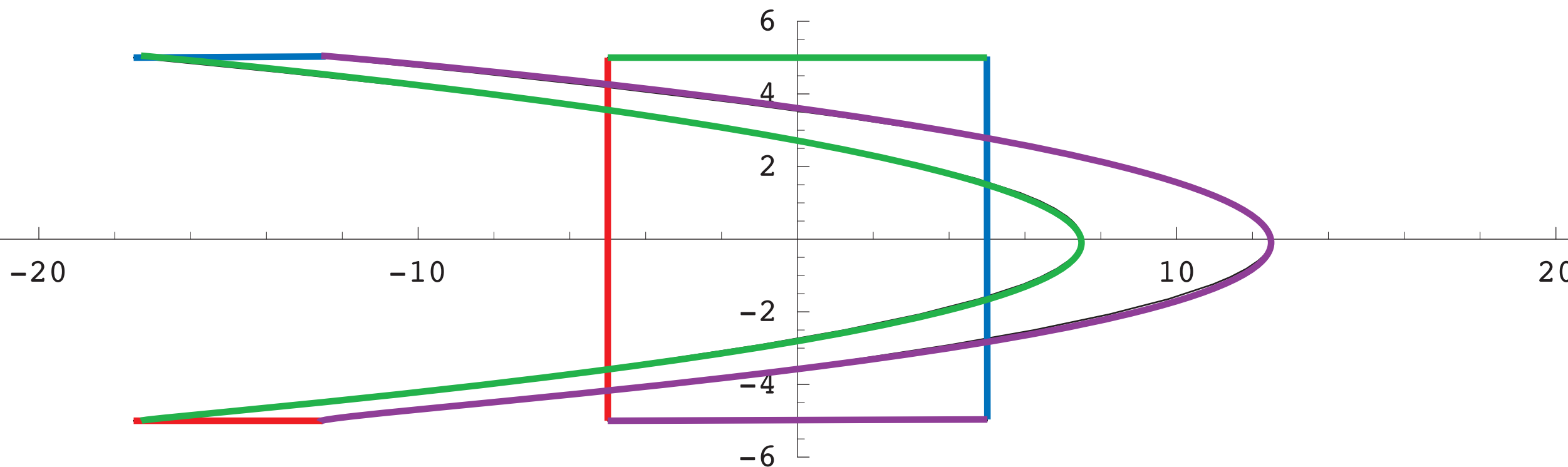
Smale horseshoe have transversal homoclinic points.



If a map $f : M \rightarrow M$ have a transversal homoclinic point, then its iteration $f^k : M \rightarrow M$ for some large k contains a Smale horseshoe (Poincare-Birkhoff-Smale Theorem)

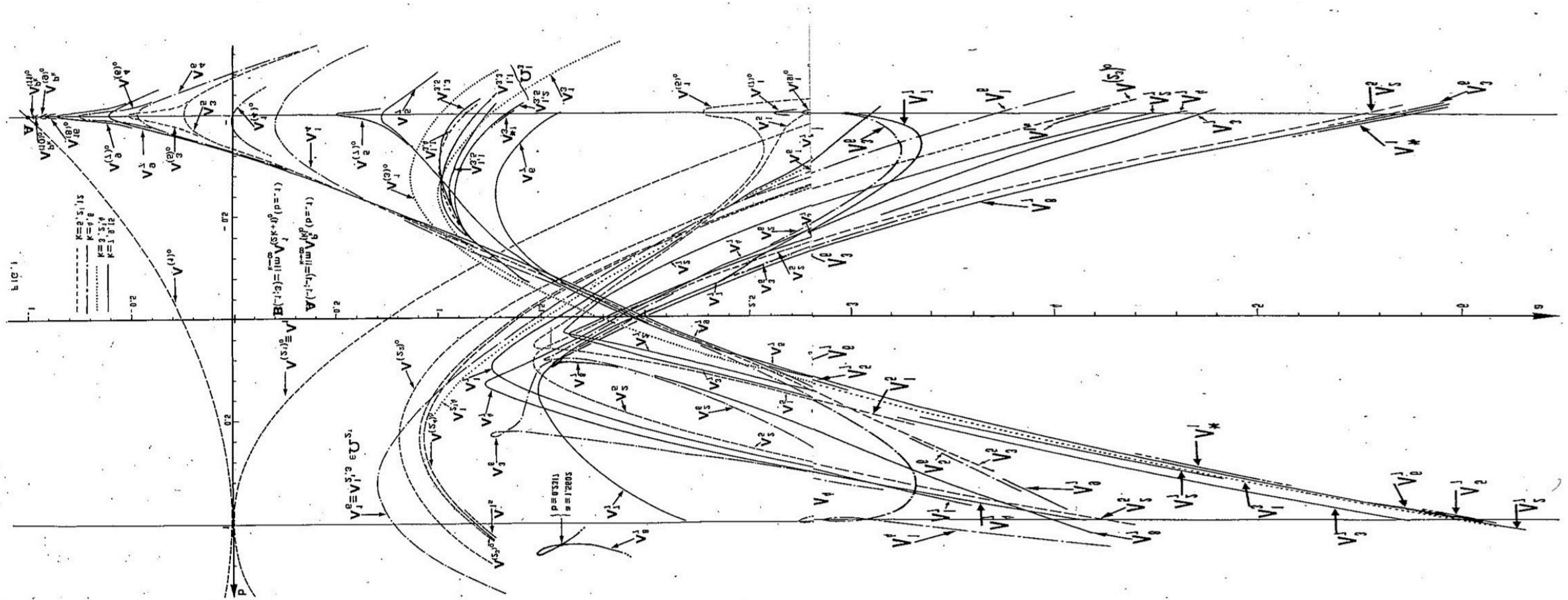
Hyperbolicity of the Hénon

[Devaney and Nitecki 1979] For any fixed b , if a is sufficiently large, then the non-wandering set of the Hénon map is uniformly hyperbolic full horseshoe.



Numerical Bifurcation Curves

A figure by Mira and El Hamouly (C.R.A.S.Paris, 1981).

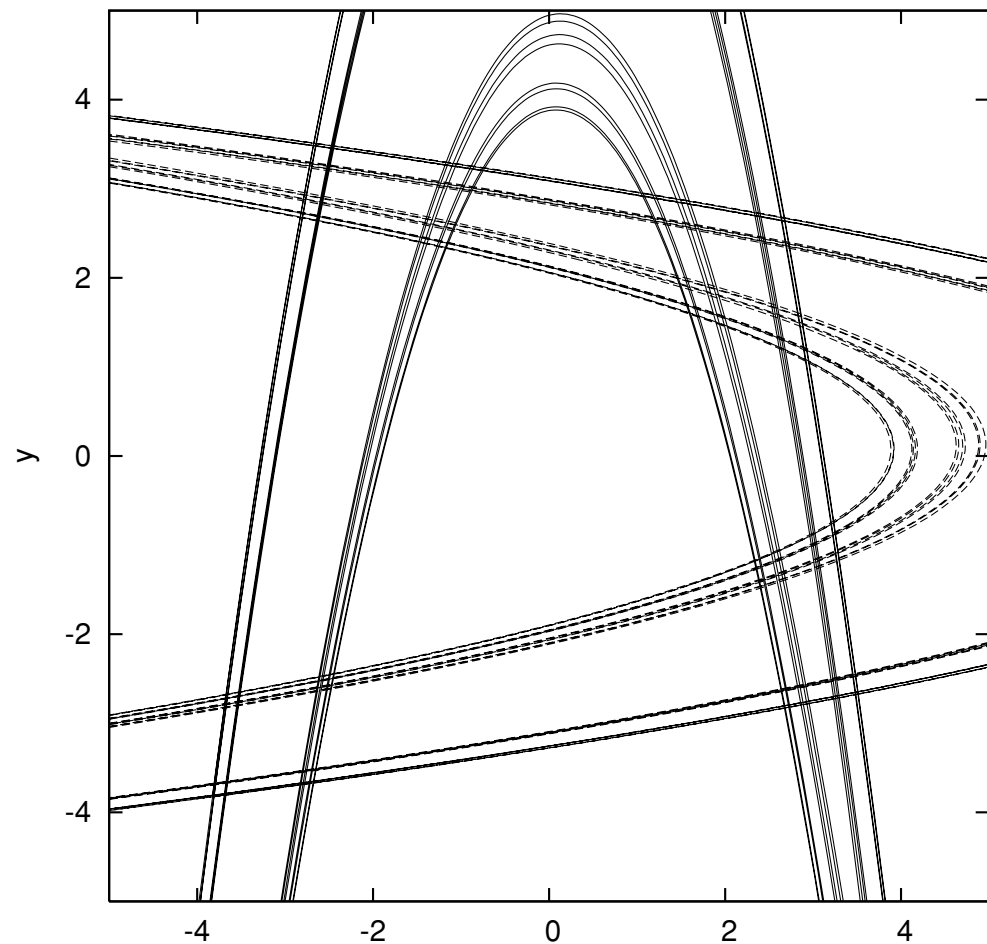


Hyperbolic Regions

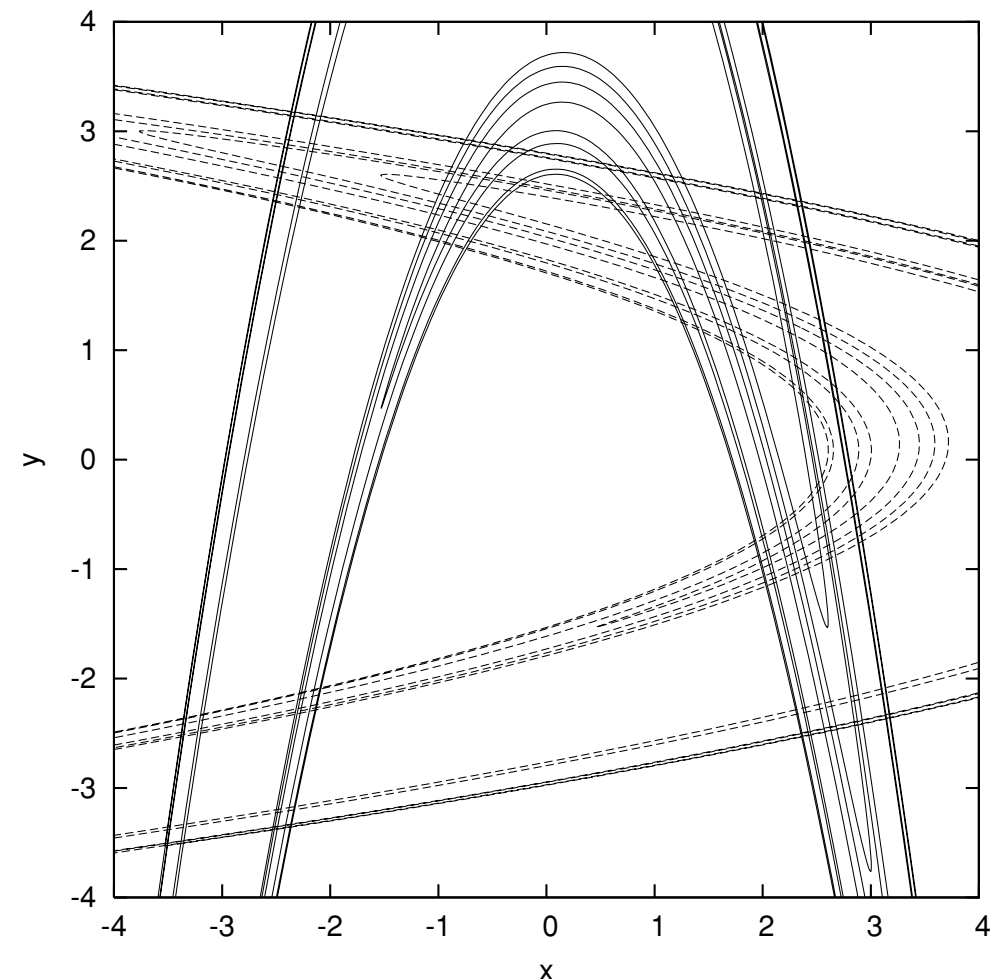
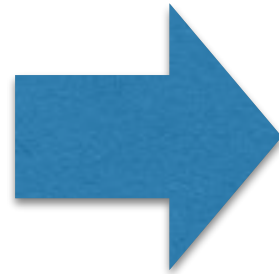


Hyperbolic parameter region of the real Hénon map (ZA, Experimental Math. 16 2007)

Unstable and Stable Manifolds



$$a = 6.0, b = 1.0$$



$$a = 5.4, b = 1.0$$

It is likely that the “first” bifurcation is a tangency between unstable and stable manifolds of the fixed pt.

First Bifurcation Problem

Problem Determine the type of “first” bifurcation and describe the dynamics at the bifurcation

Known Results for the Henon map

Ergodic Theory:

Takahashi (2014)

Real Dynamics:

Cao-Luzzatto-Rios (2008)

Complex Dynamics:

Bedford-Smillie (2006)
ZA-Ishii (2015)

The Main Result

$\mathcal{M}^\times \equiv \{(a, b) \in \mathbb{R} \times \mathbb{R}^\times : f_{a,b} \text{ attains the maximal entropy } \log 2\}$

$\mathcal{H}^\times \equiv \{(a, b) \in \mathbb{R} \times \mathbb{R}^\times : f_{a,b} \text{ is a hyperbolic horseshoe on } \mathbb{R}^2\}.$

Main Theorem (ZA & Y. Ishii 2014)

There exists an analytic function $a_{\text{tgc}} : \mathbb{R}^\times \rightarrow \mathbb{R}$ from the b -axis to the a -axis in the parameter space of $f_{a,b}$ with $\lim_{b \rightarrow 0} a_{\text{tgc}}(b) = 2$ s.t.

- $a > a_{\text{tgc}}(b)$ iff $(a, b) \in \mathcal{H}^\times$,
- $a \geq a_{\text{tgc}}(b)$ iff $(a, b) \in \mathcal{M}^\times$.

Moreover, when $a = a_{\text{tgc}}(b)$, the map $f_{a,b}$ has exactly one orbit of tangencies of the stable/unstable manifolds of certain fixed points.

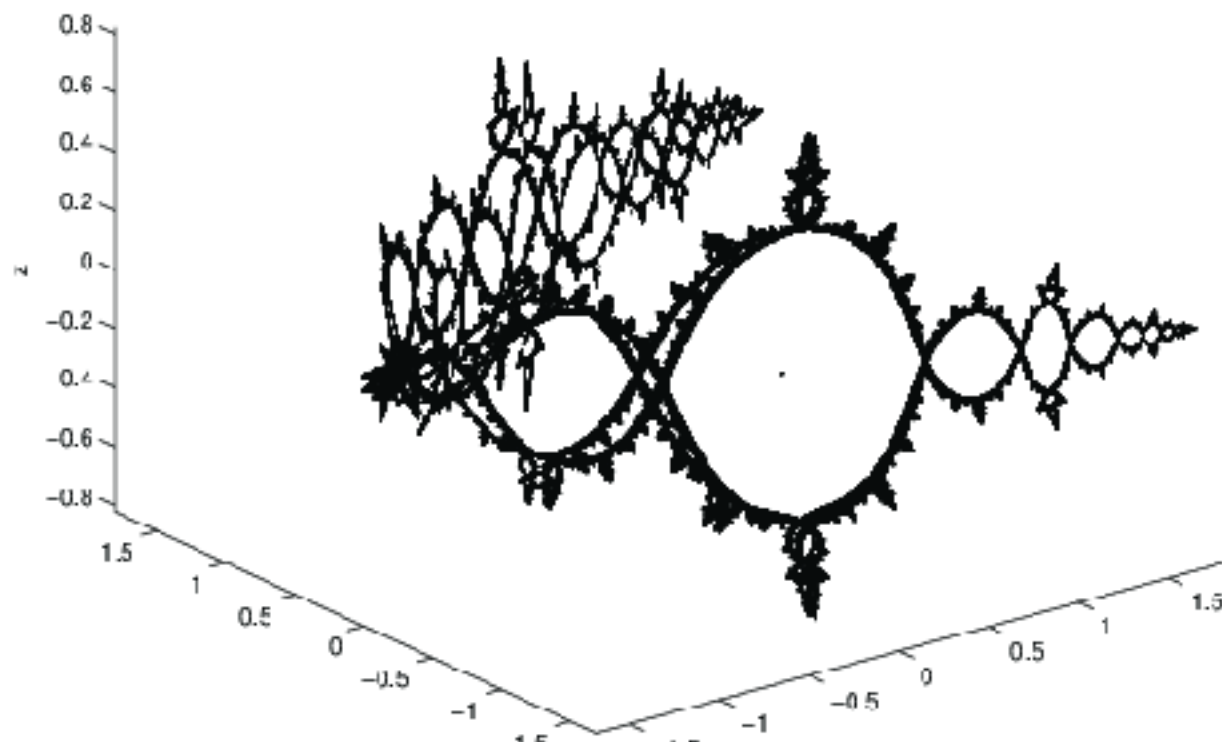
Previously shown by Bedford and Smillie (2006) for $|b| < 0.06$.

Our Strategy

Basically, we follow the proof of Bedford and Smillie:
Extend both the dynamical and parameter planes from $\mathbb{R}^2$ to $\mathbb{C}^2$ and investigate the **complexified** dynamics.

The dynamics is, however, much more complicated than Bedford–Smillie’s near-1-dim case.

Therefore, we introduce “projective box” methods and also use **rigorous numerics**.



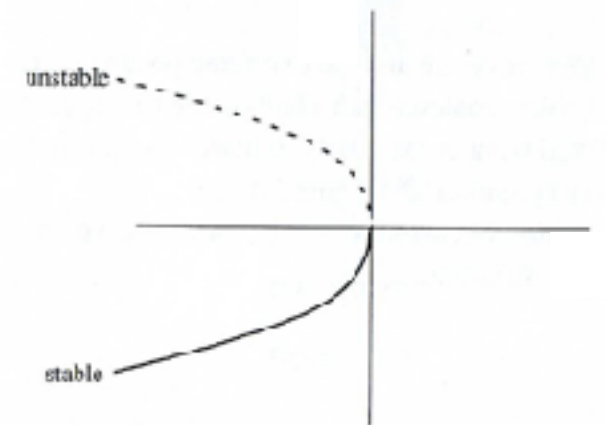
The projection of the filled Julia set of the Hénon map with $a = 1.1875$, $b = 0.15$ to $\mathbb{R}^3 = \langle \operatorname{Re} x, \operatorname{Re} y, \operatorname{Im} x \rangle$

Complexification

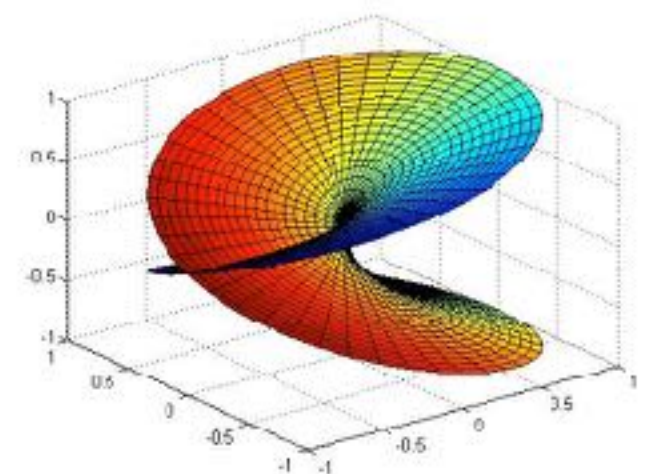
By complexification, often the dynamics becomes “robust” w.r.t. the parameters.

For example, the topological entropy, the number of periodic points (counted with multiplicity) of the complex Hénon map does NOT depend on the parameter.

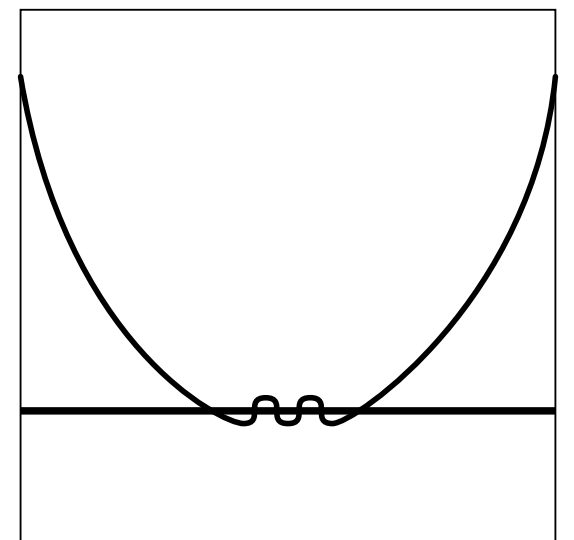
We can use complex analytic method to rule out unnecessary intersections.



real saddle node



complex saddle node



Machineries

Quasi-Trichotomy

We classify any Hénon map $f_{a,b}$ into three types (not exclusive).

Theorem (Quasi-Trichotomy)

One can construct a piecewise affine function $a_{\text{aprx}} : \mathbb{R}^\times \rightarrow \mathbb{R}$ (whose graph approximates $\partial\mathcal{M}$ and $\partial\mathcal{H}$) so that

- (1) for $(a, b) \in \mathbb{R} \times \mathbb{R}^\times$ with $a \leq a_{\text{aprx}}(b) - 0.1$, the Hénon map $f_{a,b}$ satisfies $h_{\text{top}}(f_{a,b}|_{\mathbb{R}^2}) < \log 2$,*
- (2) for $(a, b) \in \mathbb{C} \times \mathbb{R}^\times$ with $|a - a_{\text{aprx}}(b)| \leq 0.1$, the Hénon map $f_{a,b}$ satisfies the (CMC) for a family of boxes $\{\mathcal{B}_i\}_i$ in $\mathbb{C}^2$,*
- (3) for $(a, b) \in \mathbb{R} \times \mathbb{R}^\times$ with $a \geq a_{\text{aprx}}(b) + 0.1$, the Hénon map $f_{a,b}$ is a hyperbolic horseshoe on $\mathbb{R}^2$.*

Note: (2) stays valid for b in a small complex neighborhood of $\mathbb{R}^\times$.
Let $\mathcal{F}$ be the set of $(a, b) \in \mathbb{C} \times \mathbb{C}^\times$ satisfying the condition in (2).

Proof of Quasi-trichotomy (1)

Use a result of Bedford-Lyubich-Smillie (1993):

$$h_{\text{top}}(f_{a,b}|_{\mathbb{R}^2}) = \log 2 \quad \text{if and only if} \quad K_{f_{a,b}} \subset \mathbb{R}^2$$

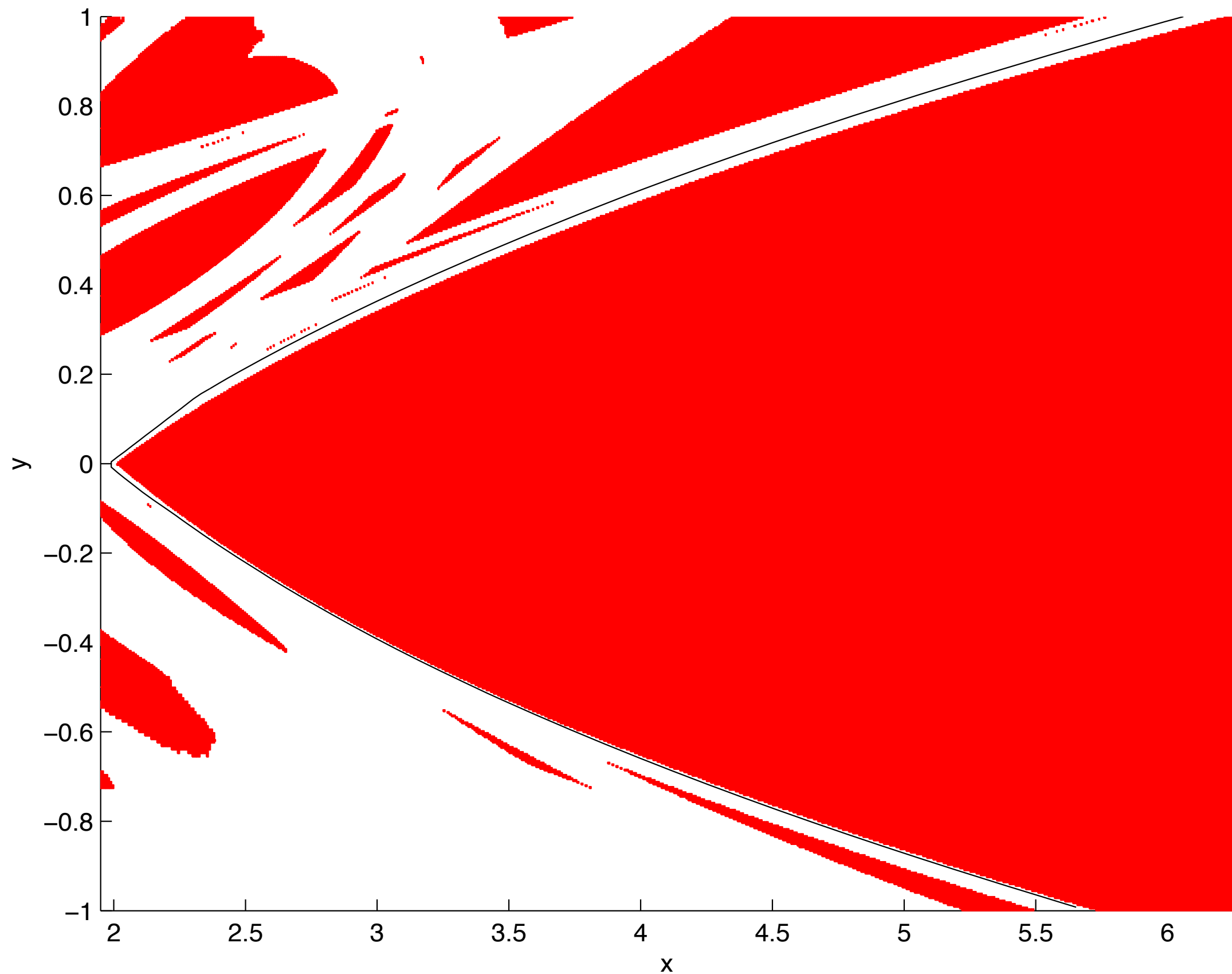
for a real Hénon map $f_{a,b}$, where K_f is the set of points in $\mathbb{C}^2$ whose forward and backward orbits by f are both bounded.

In particular, it is enough to find a saddle periodic point in $\mathbb{C}^2 \setminus \mathbb{R}^2$.

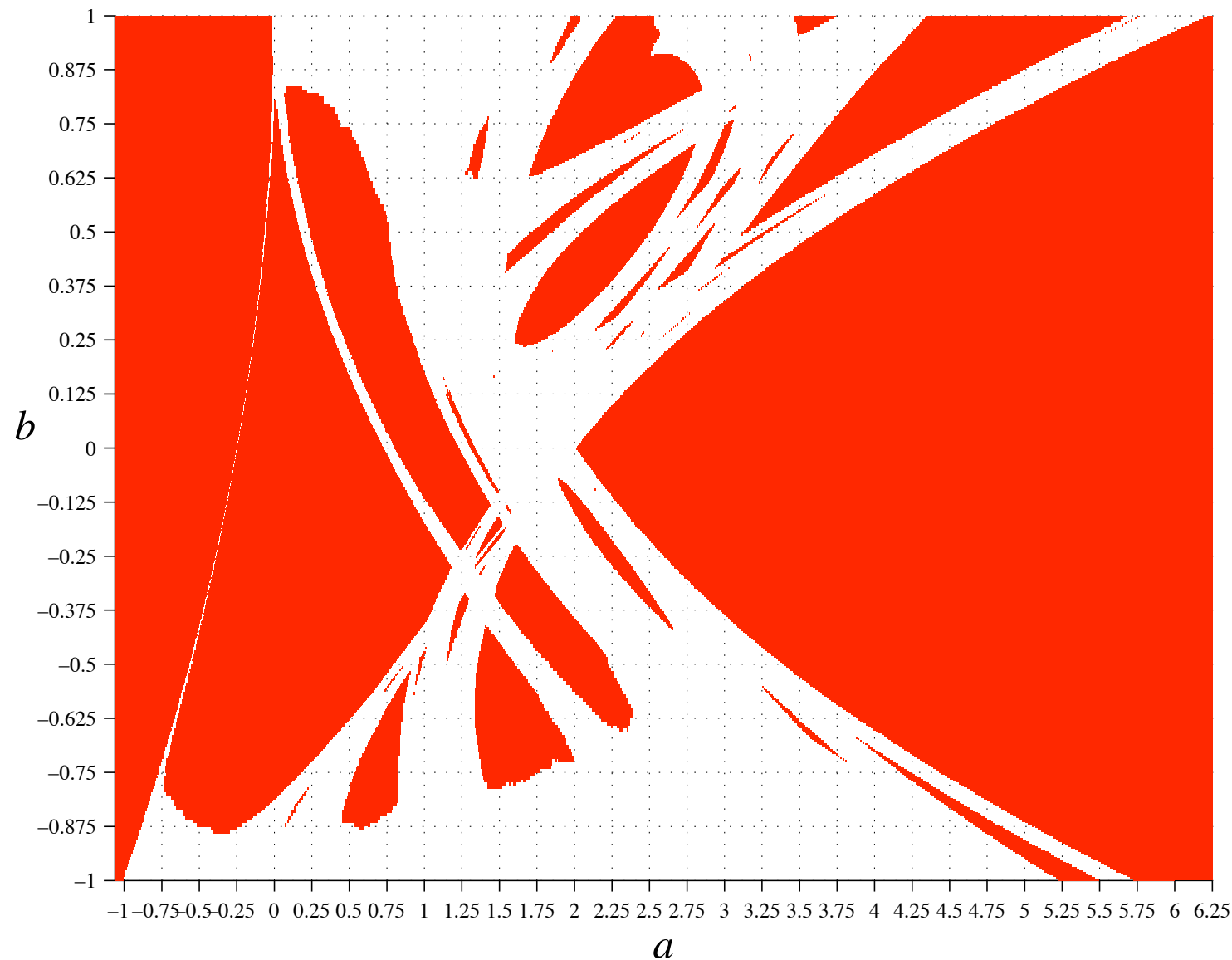
Rigorous Numerics

Using the **Interval Krawczyk Method**, we can prove the existence of such saddle point (of period 7) for all parameter values satisfying $a \leq a_{\text{aprx}}(b) - 0.1$.

SN Curves of Period 7



Proof of Quasi-trichotomy (3)



Rigorous Numerics

The uniform hyperbolicity for all parameter values satisfying $a \geq a_{\text{aprx}}(b) + 0.1$ follows from my previous work (ZA, Experimental Math 16, 2007).

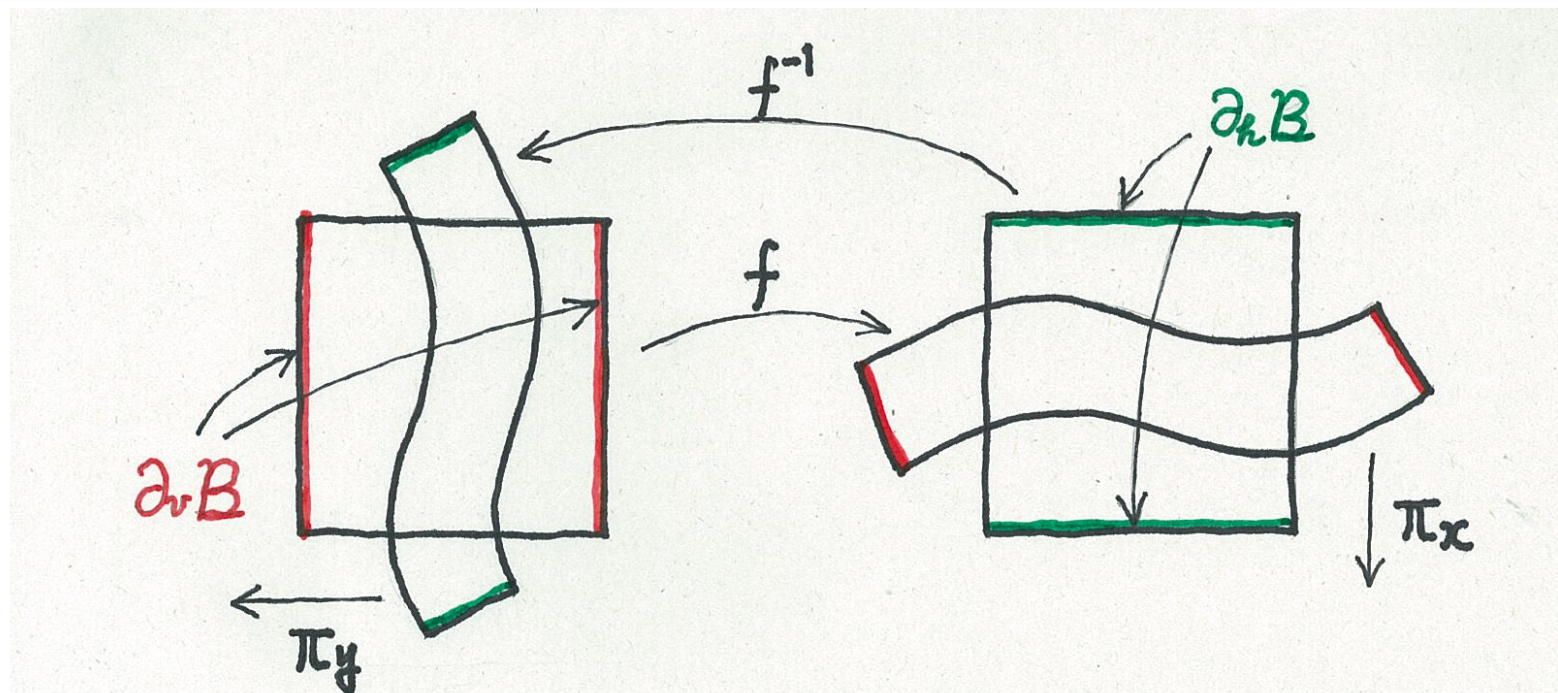
Crossed Mapping Condition

A *box* is a product set $\mathcal{B} \equiv U_x \times U_y \subset \mathbb{C}^2$, where U_x and U_y are domains in $\mathbb{C}$. Set $\partial_v \mathcal{B} \equiv \partial U_x \times U_y$ and $\partial_h \mathcal{B} \equiv U_x \times \partial U_y$.

Let $f = f_{a,b} : \mathbb{C}^2 \rightarrow \mathbb{C}^2$ be a Hénon map over $\mathbb{C}$.

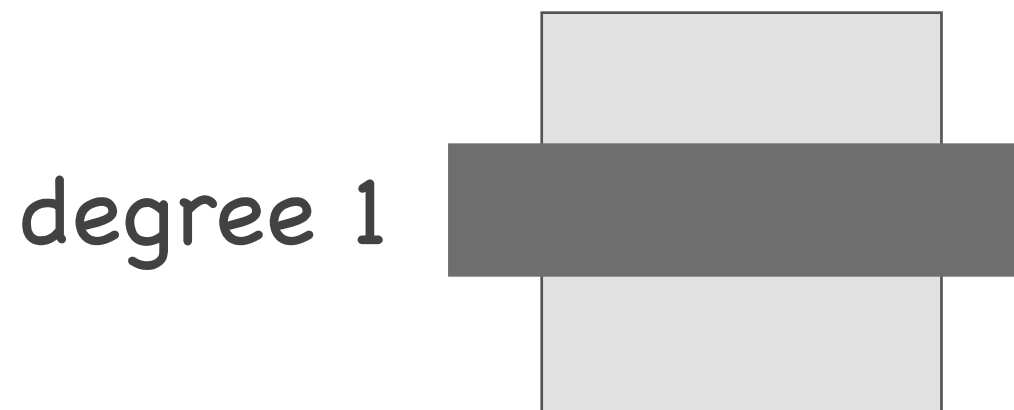
Definition (J. H. Hubbard & R. W. Oberste-Vorth)

We say $f : \mathcal{B} \rightarrow \mathcal{B}$ satisfies the *crossed mapping condition (CMC)* if $\pi_x \circ f(\partial_v \mathcal{B}) \cap U_x = \emptyset$ and $\pi_y \circ f^{-1}(\partial_h \mathcal{B}) \cap U_y = \emptyset$ hold.

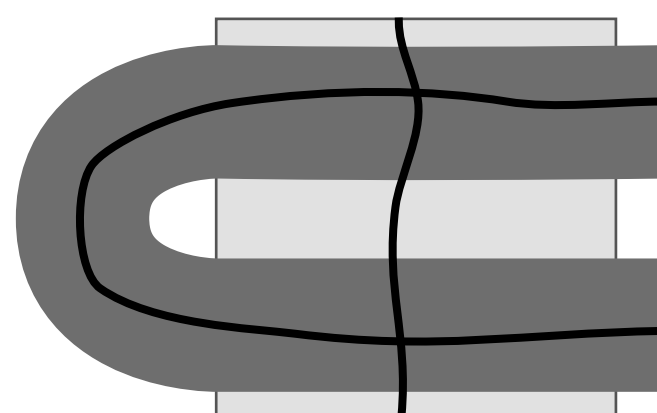
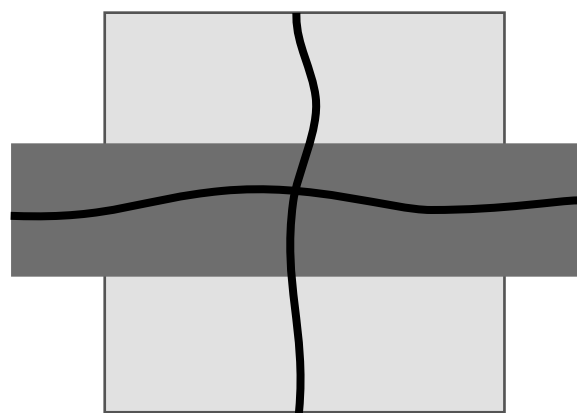


Crossed Mapping Condition

* We can define the degree of a crossed mapping

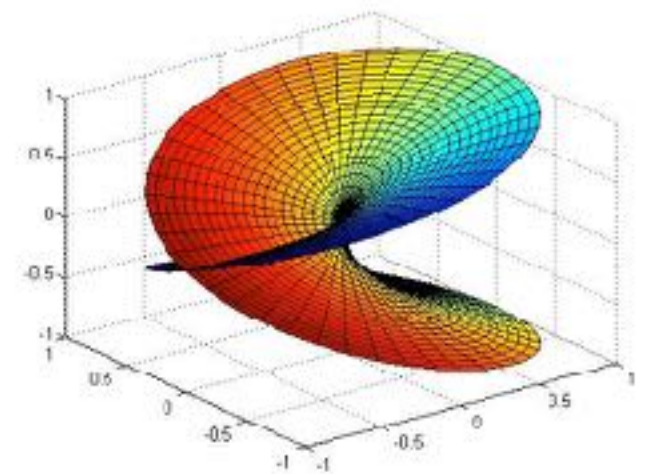
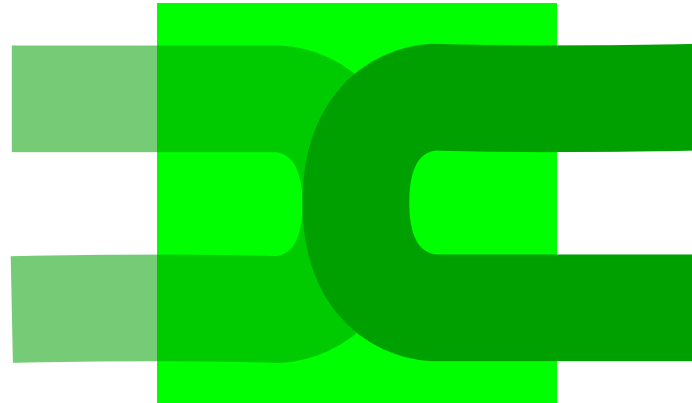


* If $f : \mathcal{B}_1 \rightarrow \mathcal{B}_2$ is a crossed mapping and D_h is a "horizontal disc" in $\mathcal{B}_1$, D_v a "vertical disc" in $\mathcal{B}_2$,
 $\#(D_v \cap f(D_h)) = \text{degree of } f \text{ (counted with multiplicity)}$

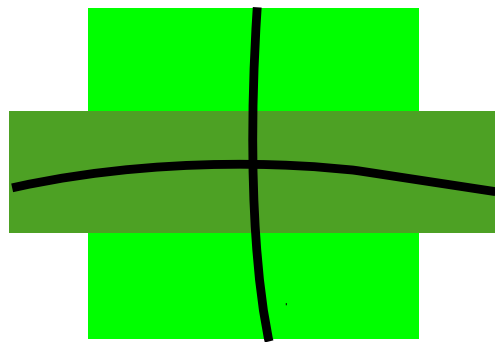


Degree and intersections

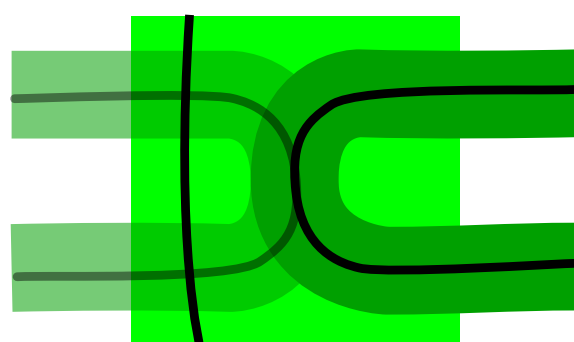
* We can define the degree of a crossed mapping
 degree 1 degree 2



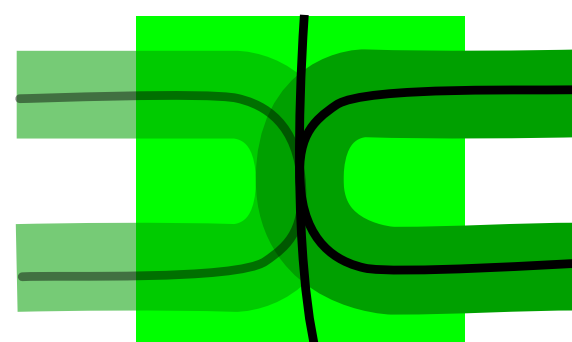
* If $f : \mathcal{B}_1 \rightarrow \mathcal{B}_2$ is a crossed mapping and D_h is a "horizontal disc" in $\mathcal{B}_1$, D_v a "vertical disc" in $\mathcal{B}_2$,
 $\#(D_v \cap f(D_h)) = \text{degree of } f \text{ (counted with multiplicity)}$



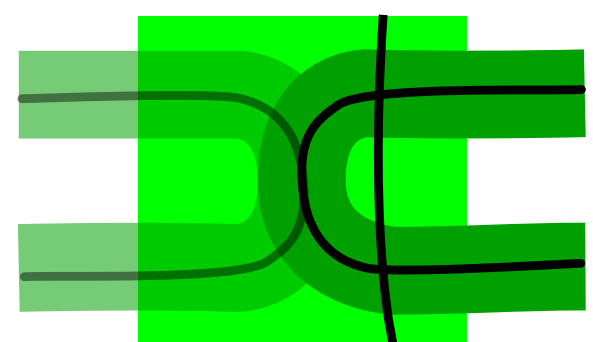
degree 1



2 imaginary intersections



tangency (multiplicity 2)



2 real intersections

Topological Hyperbolicity

Schwarz Lemma

Let $\mathbb{D} \subset \mathbb{C}$ be the unit open disk and $f : \mathbb{D} \rightarrow \mathbb{D}$ be a holomorphic map with $f(0) = 0$.

Then,

- f is a rotation about the origin, or,
- $|f'(0)| < 1$ and $f^n(z) \rightarrow 0$ for all $z \in \mathbb{D}$.

In other words,

topological hyperbolicity $\Rightarrow$ hyperbolicity

Projective Coordinates

Let $u \in \mathbb{C}^2$ and let L_u be a complex line in $\mathbb{C}^2$ such that $u \notin L_u$. Let $\pi_u : \mathbb{C}^2 \dashrightarrow L_u$ be the projection with respect to the focus u (the domain of π_u is $\mathbb{C}^2 \setminus L'_u$, where L'_u is a complex line parallel to L_u through u). We call u the *focus* of π_u .

Consider two projections $\pi_u : \mathbb{C}^2 \dashrightarrow L_u$ and $\pi_v : \mathbb{C}^2 \dashrightarrow L_v$. Take two bounded topological disks $U_u \subset L_u$ and $U_v \subset L_v$.

Lemma

Under certain assumption, $\pi_u^{-1}(U_u) \cap \pi_v^{-1}(U_v)$ is biholomorphic to a rigid bidisk in $\mathbb{C}^2$.

Below we write $U_u \times_P U_v$ for $\pi_u^{-1}(U_u) \cap \pi_v^{-1}(U_v)$ and we call it a *projective box*.

Quadrilaterals

Suppose that we are given a “not-too-bad” quadrilateral D in $\mathbb{R}^2$.

Extend two non-neighboring edges of D to obtain two lines in $\mathbb{R}^2$, whose intersection u is defined to be a focus (it is possibly ∞).

Take a complex line L_u and define a projection $\pi_u : \mathbb{C}^2 \dashrightarrow L_u$.

Similarly we obtain $\pi_v : \mathbb{C}^2 \dashrightarrow L_v$ from D .

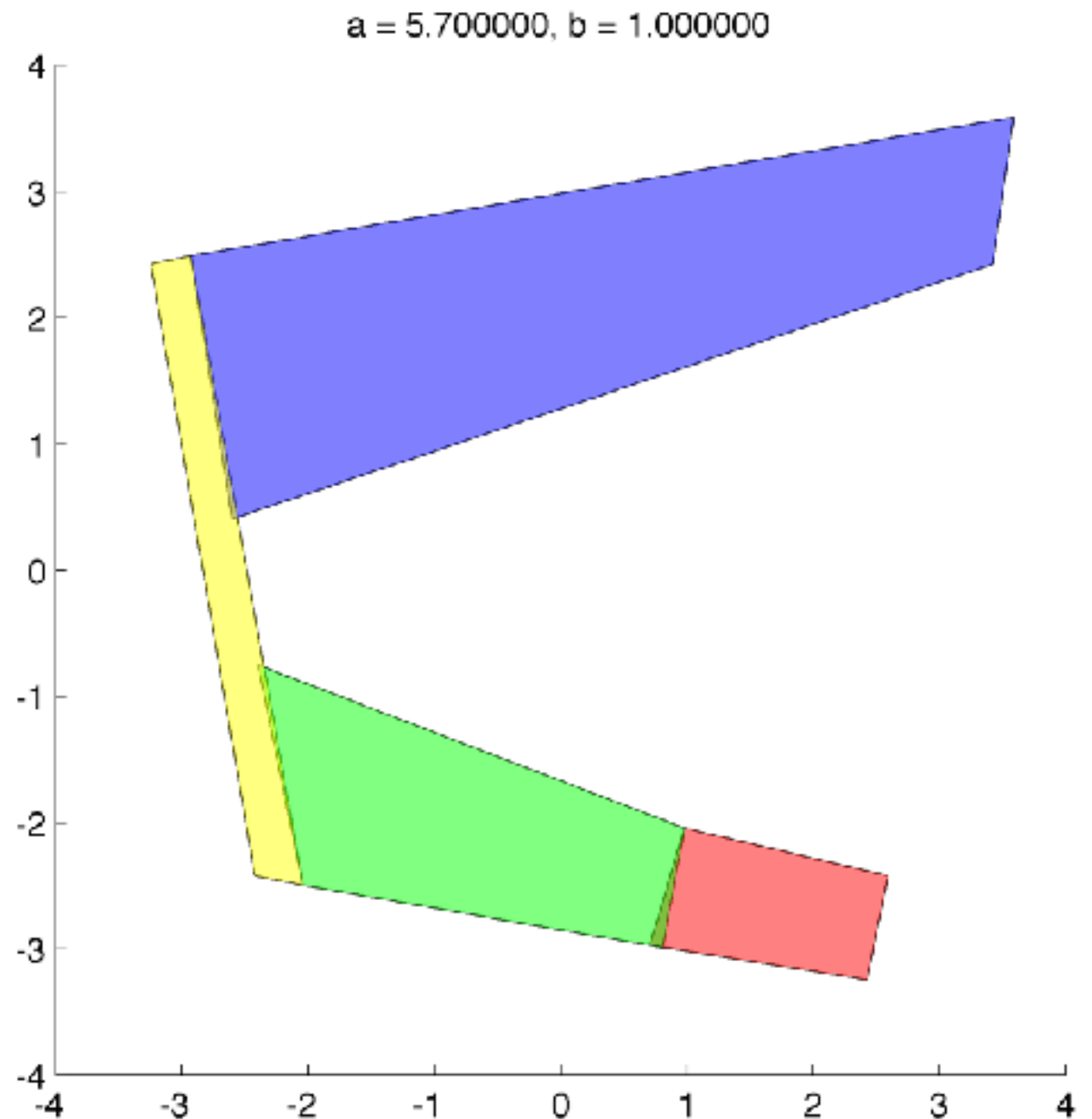
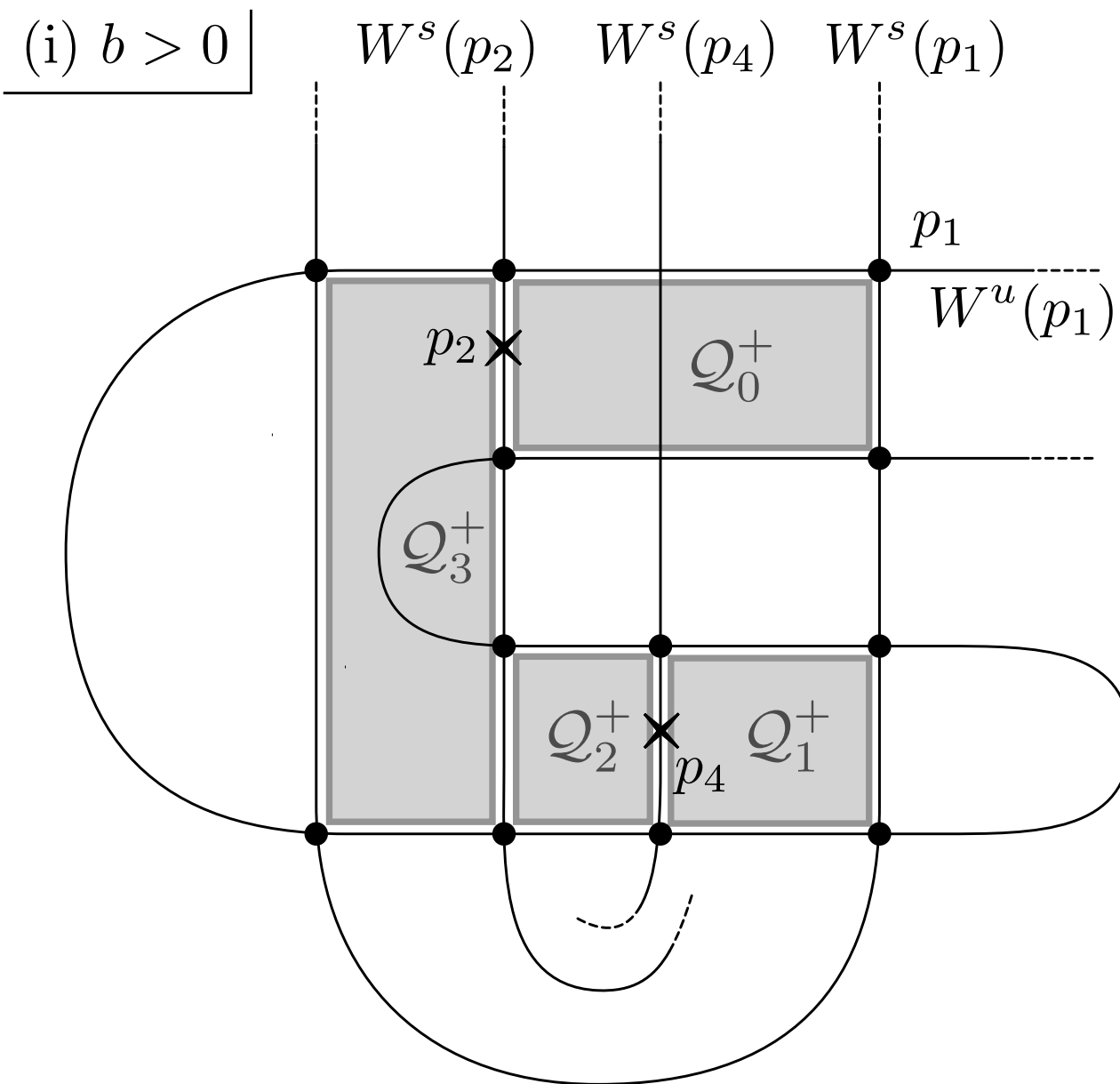
Let $I_u = \pi_u(D)$ and $I_v = \pi_v(D)$, which are intervals in $\mathbb{R} \subset \mathbb{C}$.

Choose their connected neighborhoods $U_u \supset I_u$ and $U_v \supset I_v$ and set $\mathcal{B} = U_u \times_{\mathbb{P}} U_v$ which we call a *projective box associated with the quadrilateral D* .

Real Boxes

Real Boxes

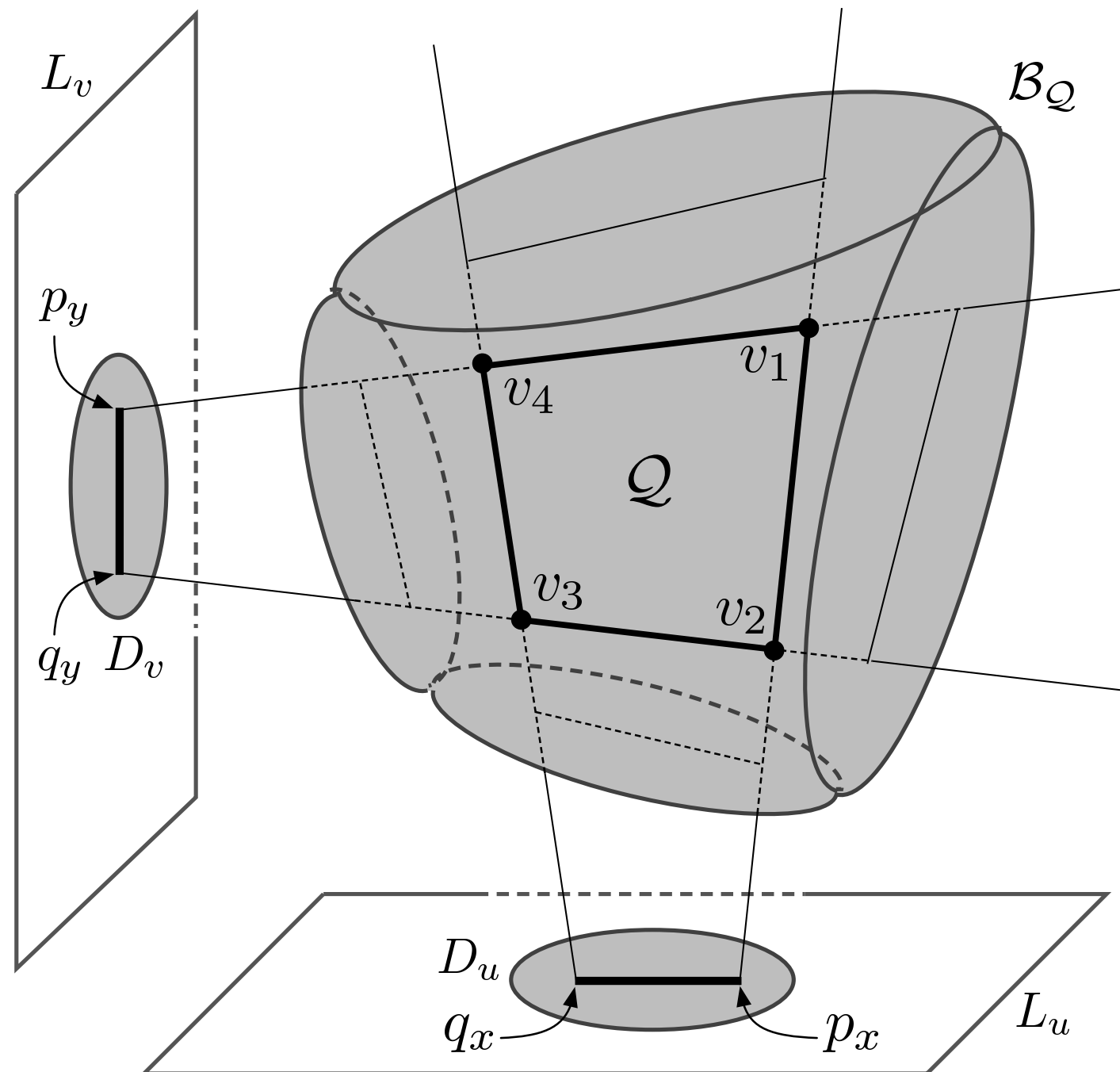
We define “real boxes” according to the trellis of the map.



Complex Boxes

Complex Boxes

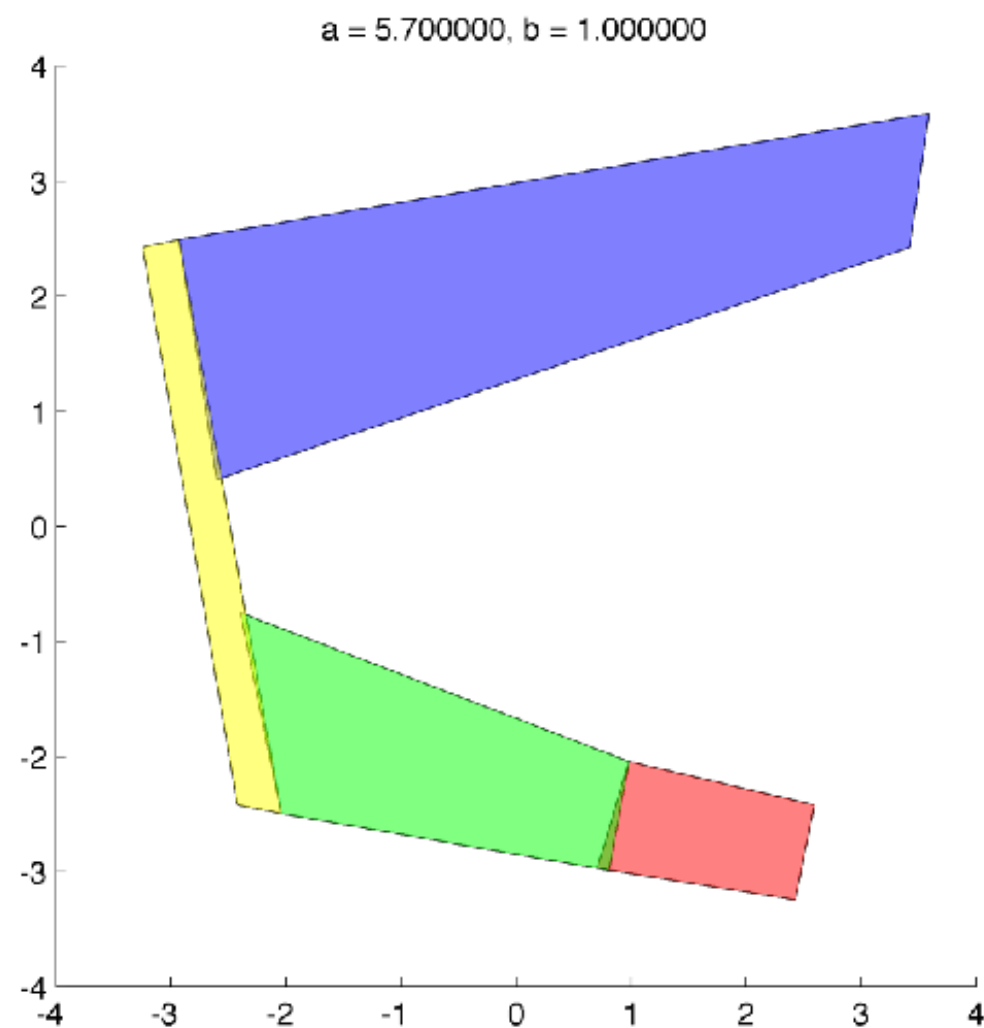
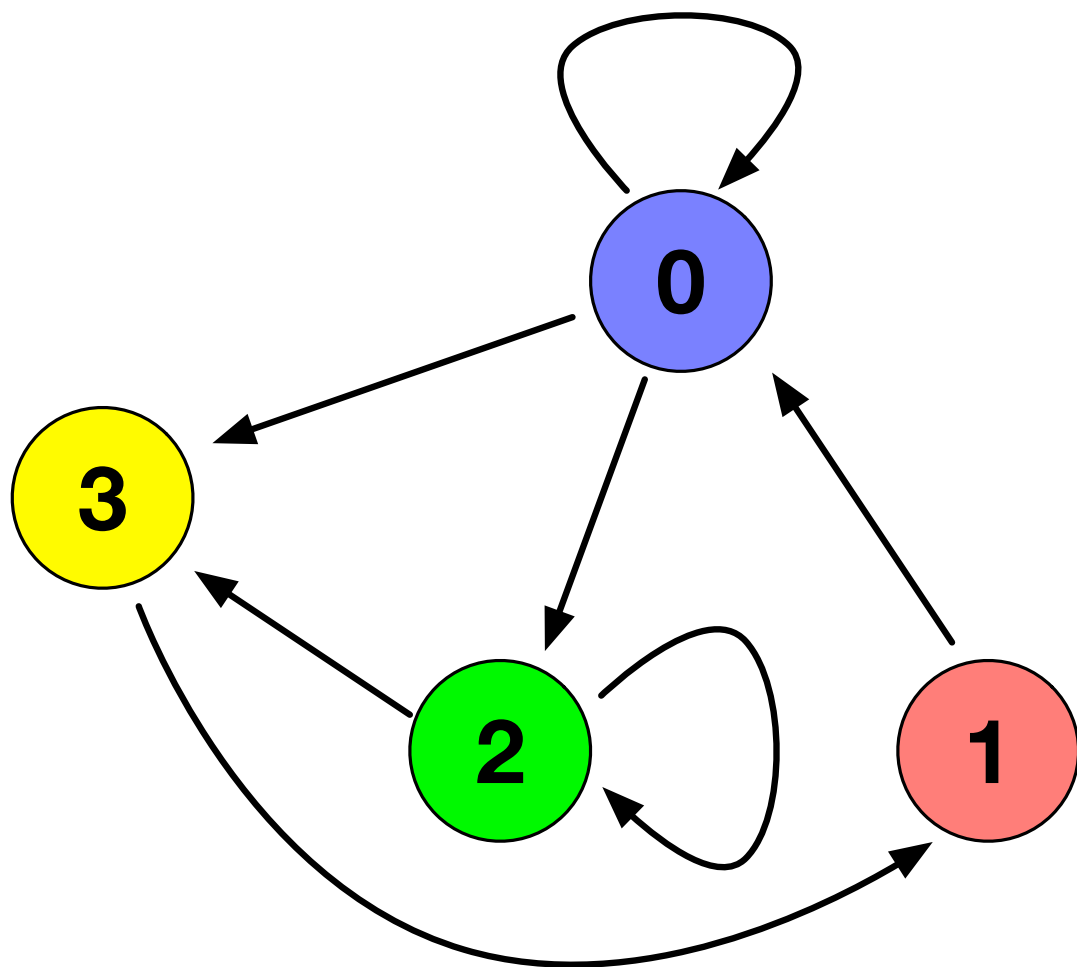
Real boxes are enlarged to complex boxes using projective coordinates.



Rigorous Numerics

Rigorous Numerics

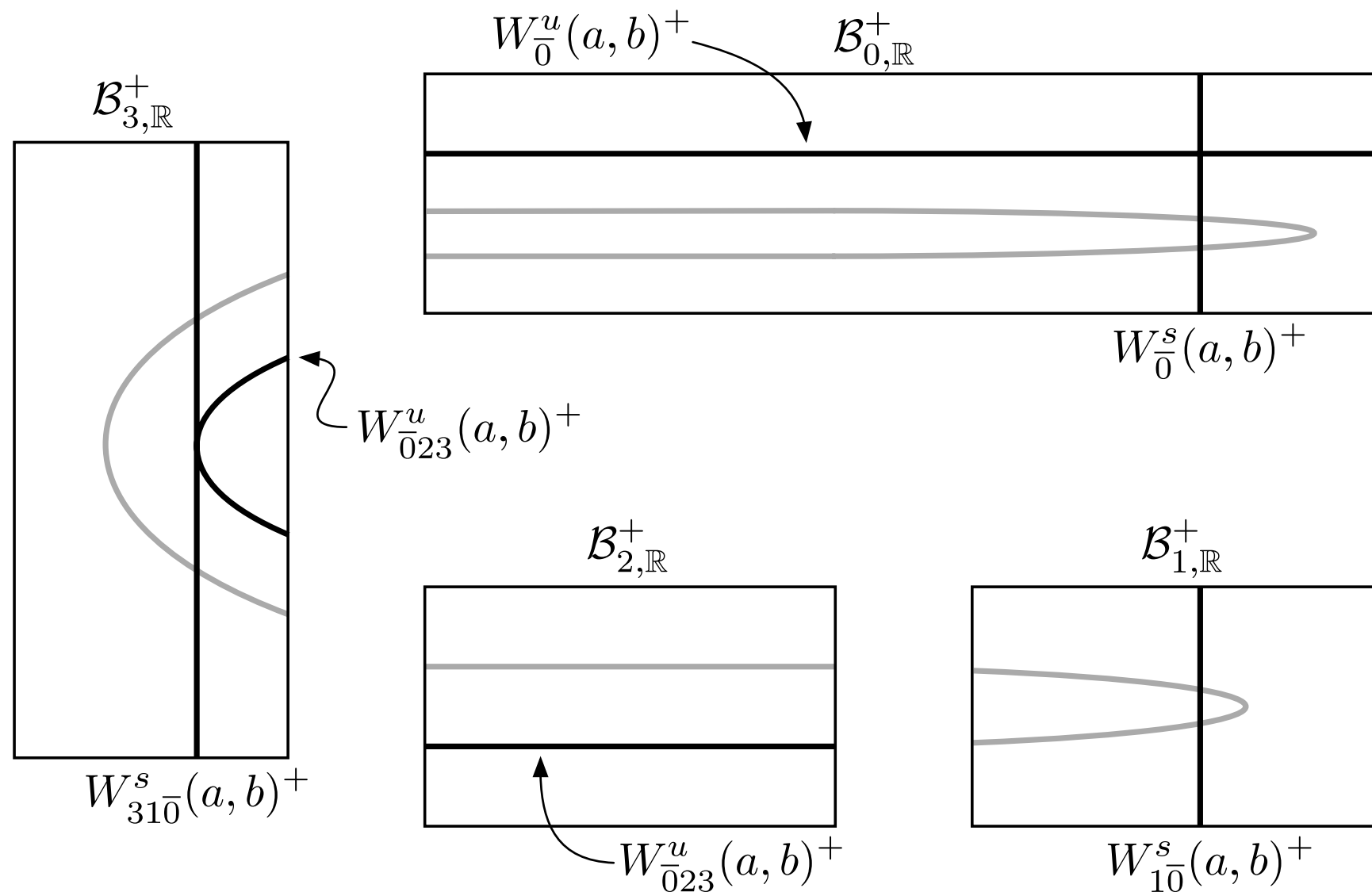
For all parameter values $|a - a_{\text{aprx}}(b)| \leq 0.1$ and all “admissible” pairs (i, j) , we check that $f_{a,b} : B_i \rightarrow B_j$ satisfies CMC.



How these machineries works

Symbolic Decomposition

Family of boxes $\{\mathcal{B}_i\}_i$ enables us to define a symbolic encoding of pieces of the invariant manifolds $V^{u/s}(p)$ in $\mathbb{C}^2$, but with overlaps.



The (CMC) in the quasi-trichotomy assures that these pieces are “nice” holomorphic disks in $\mathcal{B}_i$.

Maximal Entropy Criterion

Now we return to the real dynamics $f_{a,b} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$. Denote by $W_*^{u/s}(a, b)$ the real part of $V_*^{u/s}(a, b)$. We have the following characterization of $\mathcal{M}^\times$ in terms of “special pieces”.

Theorem (Maximal Entropy)

Suppose that $(a, b) \in \mathcal{F}_{\mathbb{R}} = \mathcal{F} \cap \mathbb{R}^2$.

- *When $b > 0$, we have*

$$\text{Card}(W_{31\bar{0}}^s(a, b) \cap W_{\bar{0}23}^u(a, b)) \geq 1 \iff (a, b) \in \mathcal{M}^\times.$$

- *When $b < 0$, we have*

$$\text{Card}(W_{41\bar{0}}^s(a, b) \cap W_{434124}^u(a, b)_{\text{inner}}) \geq 1 \iff (a, b) \in \mathcal{M}^\times.$$

Proof of Theorem (Maximal Entropy) uses the following result of Bedford-Lyubich-Smillie (1993). For a real Hénon map f , TFAE.

- $h_{\text{top}}(f|_{\mathbb{R}^2}) = \log 2$,
- if p_1 and p_2 are saddles of f , then $V^u(p_1) \cap V^s(p_2) \subset \mathbb{R}^2$.

Special Pieces

Proposition (Special Pieces)

Assume the LHS of Theorem (Maximal Entropy). Then,

- $W_{31\bar{0}}^s(a, b)$ is the left-most piece among $D_3 \cap W^s(p)$ and $W_{\bar{0}23}^u(a, b)$ is the inner-most piece among $D_3 \cap W^u(p)$ for $b > 0$.
- $W_{41\bar{0}}^s(a, b)$ is the left-most piece among $D_4 \cap W^s(p)$ and $W_{434124}^u(a, b)_{\text{inner}}$ is the inner-most piece among $D_4 \cap W^u(q)$ for $b < 0$.

This excludes the possibility of other tangencies before the first tangency of “special pieces”.

Complex Tangency Loci

The previous theorems indicates that the tangency between those pieces of invariant manifolds are responsible for the “first” bifurcation.

Definition

We define

$$\mathcal{T}^+ \equiv \{(a, b) \in \mathbb{C}^2 : V_{31\bar{0}}^s(a, b) \cap V_{023}^u(a, b) \neq \emptyset \text{ tangentially}\}$$

and

$$\mathcal{T}^- \equiv \{(a, b) \in \mathbb{C}^2 : V_{41\bar{0}}^s(a, b) \cap V_{434124}^u(a, b) \neq \emptyset \text{ tangentially}\}$$

and call them the *complex tangency loci*.

Complex Analytic Sets

A *complex analytic set* is the set of common zeros of finitely many analytic functions in $U \subset \mathbb{C}^n$. A general consideration yields that the complex tangency loci $\mathcal{T}^+$ and $\mathcal{T}^-$ form complex analytic sets, but possibly with singularities.

How to wipe out the singularities?

Lemma

Let $U_a, U_b \subset \mathbb{C}$ and let $\mathcal{T} \subset U_a \times U_b$ be a complex analytic set. If $\overline{\mathcal{T}} \cap (\partial U_a \times U_b) = \emptyset$, the projection $\pi_b : \mathcal{T} \rightarrow U_b$ is proper.

For $\mathcal{T}^\pm$, we can count the degree of π_b at $b = 0$; transversality of the quadratic family $p_a : x \mapsto x^2 - a$ at $a = 2$ yields that it is one. Proper of degree one $\implies$ complex manifold (no singularity!).

Tin Can Argument

To verify the assumption of the previous lemma, we consider

$$\mathbb{C} \supset V \xrightarrow{\varphi} V_0^u(a, b) \xrightarrow{f^2} V_{023}^u(a, b) \subset \mathcal{B}_3 \xrightarrow{\pi_u} U_3.$$

Theorem (Tin-Can)

When $b > 0$, the critical values of $\pi_u \circ f^2 \circ \varphi : V \rightarrow U_3$ are away from $\pi_u(V_{310}^s(a, b))$ for $(a, b) \in \partial_v \mathcal{F}$. Similar statement holds for the case $b < 0$ as well.

Rigorous Numerics

For all parameter values $(a, b) \in \partial_v \mathcal{F}$, we rigorously enclose the corresponding pieces of unstable and stable manifolds using **set oriented method** (GAIO-like algorithms) and check they do not intersect.

Proof of the Main Thm 1/3

Step 1. Construction of a_{tgc} .

Tin-Can $\implies$ can apply the previous lemma to show

- $\mathcal{T}^+$ is a complex submanifold (without singularities),
- the projection of $\mathcal{T}^+$ to the complex b -axis is degree one.

Let $\mathcal{T}_{\mathbb{R}}^+ \equiv \mathcal{T}^+ \cap \mathbb{R}^2$.

Symmetry of $V^{u/s}(p)$ w.r.t. complex conjugation for real $f \implies$

- $(a, b) \in \mathcal{T}_{\mathbb{R}}^+$ iff $W_{310}^s(a, b) \cap W_{023}^u(a, b) \neq \emptyset$ tangentially,
- the projection of $\mathcal{T}_{\mathbb{R}}^+$ to $\{b > 0\}$ is degree one.

Hence, $\exists \kappa^+ : \{b > 0\} \rightarrow \{a \in \mathbb{R}\}$ analytic s.t. $\text{graph}(\kappa^+) = \mathcal{T}_{\mathbb{R}}^+$,
and $\exists \kappa^- : \{b < 0\} \rightarrow \{a \in \mathbb{R}\}$ analytic s.t. $\text{graph}(\kappa^-) = \mathcal{T}_{\mathbb{R}}^-$.

Finally, define a_{tgc} to be the “join” of κ^+ and κ^- .

Proof of the Main Thm 2/3

Remark:

By Quasi-Trichotomy, it is enough to prove Main Theorem in $\mathcal{F}_{\mathbb{R}}$.

Step 2. *Proof for $\mathcal{M}^{\times}$.*

$\{(a, b) : b > 0\} \setminus \mathcal{T}_{\mathbb{R}}^{+}$ has two components, one corresponds to $\text{Card}(W_{31\bar{0}}^s(a, b) \cap W_{\bar{0}23}^u(a, b)) = 0$ and another corresponds to $\text{Card}(W_{31\bar{0}}^s(a, b) \cap W_{\bar{0}23}^u(a, b)) = 2$.

Quasi-Trichotomy + Maximal Entropy $\implies$

- $a > a_{\text{tgc}}(b)$ iff $\text{Card}(W_{31\bar{0}}^s(a, b) \cap W_{\bar{0}23}^u(a, b)) = 2$,
- $a < a_{\text{tgc}}(b)$ iff $\text{Card}(W_{31\bar{0}}^s(a, b) \cap W_{\bar{0}23}^u(a, b)) = 0$
($a \geq a_{\text{tgc}}(b)$ iff $\text{Card}(W_{31\bar{0}}^s(a, b) \cap W_{\bar{0}23}^u(a, b)) \geq 1$).

Second claim above + Maximal Entropy $\implies$ Main Theorem for $\mathcal{M}^{\times}$.

Proof of the Main Thm 3/3

Step 3. Proof for $\mathcal{H}^\times$.

Since we have proved $\mathcal{M}^\times = \{a \geq a_{\text{tgc}}(b)\}$ and $\{a > a_{\text{tgc}}(b)\}$ is open, it follows that $\mathcal{H}^\times \subset \{a > a_{\text{tgc}}(b)\}$.

Let us prove $\supset$. Take $(a, b) \in \{a > a_{\text{tgc}}(b)\}$. By the first claim in Step 2, $\{a > a_{\text{tgc}}(b)\} = \{\text{Card}(W_{31\bar{0}}^s(a, b) \cap W_{\bar{0}23}^u(a, b)) = 2\}$.

$\implies$

- $h_{\text{top}}(f_{a,b}|_{\mathbb{R}^2}) = \log 2$ (by Maximal Entropy).
- no tangency between $W^u(p)$ and $W^s(p)$ (by Special Pieces).

$\implies$ uniform hyperbolicity of $f_{a,b}$ (by Bedford-Smillie).

$\{a > a_{\text{tgc}}(b)\} \cap \mathcal{H}^\times \neq \emptyset$ + structural stability of $f_{a,b}$ + connectivity
 $\implies f_{a,b}$ is a hyperbolic horseshoe on $\mathbb{R}^2$. Q.E.D.